Clinical Localization of Stroke

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KEYWORDS

• Acute stroke • Clinical localization • Arterial vascular territory • Stroke assessment
• National Institutes of Health Stroke Scale

KEY POINTS

- Expert stroke nursing care is supported by knowledge of vascular anatomy in relation to brain parenchymal physiology.
- Clinical localization is the process by which stroke clinical findings are associated with affected arterial territories in the brain.
- Although clinical localization is rarely taught to nonneurologists, nurses caring for acute stroke patients must master aspects of the localization process to enable rapid diagnosis and treatment and facilitate evidence-based ongoing evaluation.

INTRODUCTION

Clinical localization is the process of aligning stroke symptoms with specific vascular territories in the brain.[1–5] Although noncontrast computed tomography (CT) is essential to rule out hemorrhage, patients early into ischemic stroke (<6 hours) often present with normal CT scans; therefore, ischemic stroke is a clinical diagnosis dependent on localization of symptoms to specific arterial distributions. Localization is rarely taught to nonneurologists; it is a skill that requires knowledge of brain anatomy and physiology in relation to the arterial vasculature.[1,5] However, nurses intent on developing expertise in acute stroke are well served to understand at least the most classic stroke findings associated with arterial clinical localization to support their understanding of stroke progression or improvement, and to assist in prognostication of patient outcomes.

CLINICAL LOCALIZATION OF ACUTE STROKE

The brain is made up of discreet sections that each have unique functions; when injured, these areas produce disabilities that examiners identify in symptom clusters

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representative of discreet arterial brain regions. Localization is dependent on an understanding of the physiology within each area of the brain alongside the vasculature associated with these regions so that stroke arterial distribution is identified.[1,4]

Brodmann Classification and Cortical Arterial Distribution

In the early 1900s, Brodmann[2] mapped what is called the "cytoarchitecture of the brain" (**Fig. 1**), a numbered system classifying cerebral cortical physiology in relation to autopsy findings demonstrating infarction or injury. Although the system is incomplete, Brodmann Classification remains the classic model informing practitioners today about discreet areas within the brain that are responsible for important cortical functions. Stroke clinical localization combines knowledge of Brodmann classification with an understanding of the vasculature that serves these regions, so that symptoms suggesting abnormal function may be associated with reduced blood flow in a discreet portion of an artery.[1]

The brain's arterial distribution is divided into anterior and posterior divisions (**Fig. 2**). The majority blood flow is directed to the anterior circulation of the brain, which is derived from the internal carotid arteries (ICA). In approximately 70% of individuals, the ICAs arise from the common carotid arteries (CCA) bilaterally, with the right CCA arising from the innominate artery and left CCA arising directly from the aortic arch. However, the most common arch variant occurring in about 13% of individuals is the "bovine aortic arch," where the left common carotid shares its origin with the right CCA in the innominate artery or it arises from the innominate artery more distally than the right CCA. Bovine aortic arch presentations are more common among blacks than whites, although the reason is unknown. Regardless of its origin, the CCA bifurcates into the external carotid artery, which goes on to supply the face and other

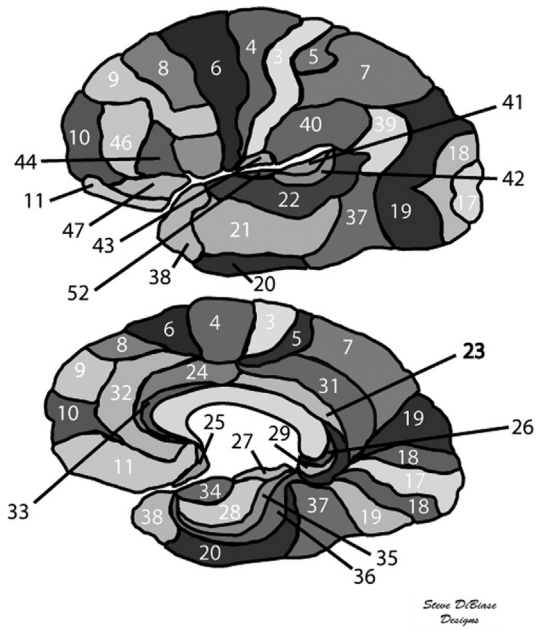


Fig. 1. Brodmann cytoarchitecture of the brain. (*Reprinted* with permission, Health Outcomes Institute, LLC, Fountain Hills, Arizona, USA.)

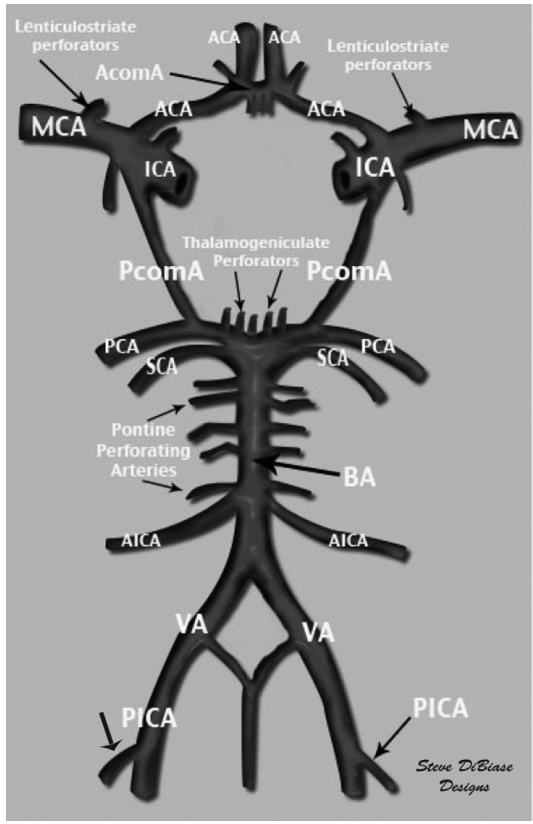


Fig. 2. Arterial vasculature of the brain. (*Reprinted* with permission, Health Outcomes Institute, LLC, Fountain Hills, Arizona, USA.)

head structures, and the ICAs, which travels deep within the brain, ultimately giving off the ophthalmic arteries (OA), the 2 posterior communicating arteries (PcomA), the anterior cerebral arteries (ACA), and the middle cerebral arteries (MCA). The posterior circulation of the brain is derived from the vertebral arteries (VA). The right VA arises from the innominate artery, which then goes on to become the right subclavian artery, and the left subclavian arises from the aortic arch and gives off the left VA. The VAs travel through the vertebral column winding their way up the neck into the skull, where they give off the posterior inferior cerebellar arteries (PICA) and the anterior spinal arteries. At the level of the pons, the VAs fuse to form the single basilar artery (BA), which ultimately gives off the anterior inferior cerebellar arteries (AICA), the superior cerebellar arteries (SCA), and at the most distal aspect of the BA, the posterior cerebral arteries (PCA). Both the anterior and the posterior circuits connect deep within the brain to form the circle of Willis.[3,4]

The 3 main pairs of arteries supplying the cerebral cortex are the MCAs (largest distribution covering the lateral and inferior aspects of the frontal, parietal and temporal lobes), the ACAs (covering the anterior and superior-medial aspects of the frontal lobes, and the superior-medial aspects of the parietal lobes), and the PCAs (covering the occipital lobe, lateral-posterior-inferior aspects of the parietal lobes, as well as the thalamus, and the superior aspects of the brainstem and cerebellum).[3] **Fig. 3**

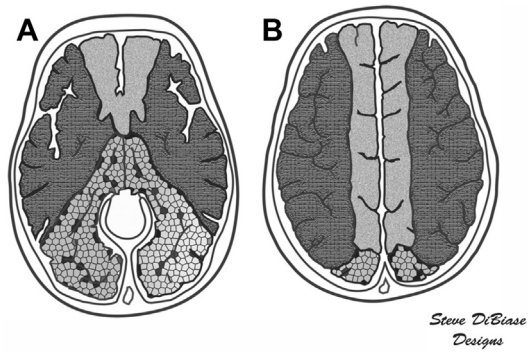


Fig. 3. Cortical arterial distribution. (*A*) View from the inferior base of the cortex. (*B*) Superior axial view of the cortex. Dark gray = MCA distribution; light gray = ACA distribution; speckled gray = PCA distribution. (*Reprinted* with permission, Health Outcomes Institute, LLC, Fountain Hills, Arizona, USA.)

illustrates cortical arterial distribution for the MCA, ACA, and PCA bilaterally; knowledge of this arterial layout is essential to support an understanding of cortical stroke clinical localization.[1] The first portions of the MCA, ACA, and PCA arteries before any bifurcation are commonly called the M1, A1, or P1 segments, respectively, whereas the segments after bifurcations are commonly referred to as M2, A2, or P2 segments; numbering of each of these divisions continues as the arterial distribution progresses further from their point of origin. The single anterior communicating artery (AcomA) is situated at the A1-A2 bifurcation connecting the right and left ACAs, whereas the 2 PcomAs, which originate from the ICAs bilaterally, move posterior, ultimately connecting at the P1-P2 bifurcation of the PCAs.[3] Although estimates vary depending on reports, as many as 50% of patient samples have been found to lack a "complete" circle of Willis; most commonly these anomalies involve one of the following[5]:

- A missing single A1 segment of the ACA with the remaining A1 segment supplying blood to the ipsilateral A2 segment as well as the contralateral A2 ACA segment via the AcomA;
- A missing P1 segment of the PCA with the PcomA supplying blood flow to the P2 PCA segment (referred to as "fetal origin of the PCA");
- A missing PcomA on 1 side.

Relating Neurologic Findings to Arterial Distribution

A complete neurologic assessment is essential in all acute stroke patients. It must be performed quickly to collect assessment data and to assign specific suspect arterial territories in under a minute's time, to decide if findings do not follow a discreet territory, such as large intracranial hemorrhages that may cross arterial distributions, or to identify findings that are highly suspicious for a large number of stroke mimic diagnoses. **Table 1** presents findings aligned by vascular territories, anatomic structures, and assessment items on the National Institutes of Health Stroke Scale (NIHSS).

The NIHSS (**Table 2**) was originally developed for clinical trials of reperfusion therapies in acute ischemic stroke[6] and provides a structured method to tailor the neurologic assessment in a manner that allows practitioners to collect pertinent data and assign a score.[7–9] The NIHSS has a bias for left hemisphere MCA territory stroke in

Table 1
Clinical Localization of Stroke

Lobe/Location	Function	Circulation	Neuro-vascular Territory	Common Pathologic Findings
Cerebral Cortex (Telencephalon)				
Frontal Lobe	Voluntary motor function *Brodmann's Area:4*	Anterior	ACA (Superior-medial aspect)	• Leg Weakness *NIHSS Item: 6*
			MCA (lateral and inferior aspects)	• Facial weakness – *Central Cranial Nerve VII* *NIHSS Item: 4* • Dysarthria (slurred speech) *NIHSS Item: 10* • Dysphagia (difficulty swallowing) *Central IX & X Nerves* • Arm weakness *NIHSS Item: 5*
	Broca's area *Brodmann's Area: 44* • Expressive language (ability to name objects)		MCA (usually left)	• Expressive Aphasia (inability to express language) *NIHSS Item: 9*
	Cognitive function *Brodmann's Areas:9–11* • Orientation • Memory • Judgment • Arithmetic • Abstract Thinking		ACA	• Orientation *NIHSS Item: 1B&1 C; influenced by NIHSS Item 9*
Parietal Lobe	Primary somesthetic sensory assessment *Brodmann's Areas: 1–3* • Pinprick • Vibration • Position sense • Temperature • Touch		ACA (Superior-medial aspect) MCA (lateral and inferior aspects)	• Tactile Sensation (discrimination of pinprick) *NIHSS Item: 8*

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Table 1
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Lobe/Location	Function	Circulation	Neuro-vascular Territory	Common Pathologic Findings
	Somesthetic association of external stimuli spatial attention *Brodmann's Areas: 5,7* • Double simultaneous stimulation • Stereognosis • Graphesthesia		MCA (usually right)	• Neglect assessment (double simultaneous stimulation, tactile and visual) *NIHSS Item: 11*
	Wernicke's Area *Brodmann's Areas: 37,39,40* • Receptive language		MCA (usually left)	• Receptive (Wernicke's or fluent) aphasia *NIHSS Item: 9*
Temporal lobe	Primary Auditory *Brodmann's Area: 41*		MCA	• Hearing loss; uncommon finding, generally not tested unless evident with sudden onset; central cranial nerve VIII *Not represented on NIHSS*
Occipital lobe	Primary Visual *Brodmann's Area: 17* Visual Association *Brodmann's Area: 18*	Posterior	PCA – Cortical vision MCA – Visual pathways OA – monocular blindness	• Vision; peripheral visual fields; differentiation between monocular and binocular lesions *NIHSS Item: 3*
Subcortex				
Anterior, lateral and inferior white matter tracks	• Motor relay • Sensory relay	Anterior	ACA MCA Lenticulo-striate perforators	• Pure motor loss *NIHSS Items 4–6* • Pure sensory loss *NIHSS Item 8* • Mixed motor and sensory loss *NIHSS Items: 4–6 and 8*

Basal nuclei	• Symmetric rhythmic movement • Suppresses muscle tone • Influences postural adjustments	Anterior	MCA Lenticulo-striate perforators	• Pure motor deficit *NIHSS: 5,6*
Posterior-lateral white matter tracks	• Motor relay • Sensory relay	Posterior	PCA Thalamo-geniculate perforators BA	• Pure motor loss *NIHSS Items 4–6* • Pure sensory loss *NIHSS Item 8* • Mixed motor and sensory loss *NIHSS Items: 4–6 and 8*
Thalami	• Relay of all sensory transmission except olfaction	Posterior	PCA Thalamo-geniculate perforators	• Thalamic pain syndrome *NIHSS Item 8* • Pure sensory loss *NIHSS Item 8*
Cerebellum				
Neo-cerebellum (middle vermis, cerebellar hemisphere)	• Projects fibers to the cerebral cortex through the thalamus; plays a role in planning and initiation of movements and regulating fine limb movement	Posterior	BA SCA AICA	• Ataxia and gait disturbances *NIHSS Item: 7* • Dysarthria (slurred speech) *NIHSS Item: 10*

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Table 1
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Lobe/Location	Function	Circulation	Neuro-vascular Territory	Common Pathologic Findings
Archi-cerebellum (Flocculo-nodular lobe)	• Processes impulses from the vestibular apparatus of the inner ear to maintain equilibrium, balance and posture • Multiple connections with the ocular centers to facilitate eye movement in response to postural changes	Posterior	BA AICA	• Vertigo, nystagmus *Not represented on NIHSS*
Paleo-cerebellum (anterior lobe vermis, paramids, uvula, para-flocculus)	• Referred to as the spinocerebellum because it receives input primarily from the spinal cord; plays a role in control of muscle tone and axial limb movements	Posterior	PICA BA AICA	• Ataxia and gait disturbances *NIHSS Item: 7* • Dysarthria (slurred speech) *NIHSS Item: 10*
		Brainstem		
Midbrain (Mesen-cephalon)	• Cell bodies for CN III & IV ○ Pupillary constriction ○ Eye movement • Level of consciousness (ascending reticular formation) • Ascending sensory tracts • Descending motor tracts	Posterior	PCA PcomA Thalamo-geniculate perforators Anterior and posterior choroidal arteries SCA BA	• Pupillary dilation *Not represented on NIHSS* • EOM dysfunction *NIHSS Item: 2* • Diplopia *Not represented on NIHSS* • Top of the basilar syndrome with cortical blindness *NIHSS Item: 3* • Decreased LOC *NIHSS Item: 1* • Motor/sensory pathway disruption *NIHSS Items: 4,5,6,8*

Pons (Meten-cephalon)	• Cell bodies for CN V-VIII ○ Facial sensation & jaw clench ○ Eye movement ○ Facial expression ○ Auditory reception & balance/equilibrium • Pneumotaxic center • Apneustic center • Level of consciousness (ascending reticular formation) • Ascending sensory tracts • Descending motor tracts	Posterior	SCA Pontine paramedian perforators AICA Internal auditory artery BA	• Facial sensation *NIHSS Item: 8* • Gaze palsies *NIHSS Item: 2* • Diplopia *Not represented on NIHSS* • Facial droop *NIHSS Item: 4* Note: *May* present with bilateral symptoms, whereby the facial droop is ipsilateral to the pontine infarction and the limb weakness is contralateral to the infarction • Hearing loss; cranial nerve VIII; uncommon finding, generally not tested unless evident with sudden onset Not represented on NIHSS • Decreased LOC *NIHSS Item: 1* ○ Locked-in syndrome • Motor/sensory pathway disruption *NIHSS Items: 4,5,6,8* • Respiratory arrest or insufficiency *Not represented on NIHSS*
Medulla Oblongata (Mylen-cephalon)	• Cell bodies of CN IX-XII • Cardiac center • Vasomotor center • Respiratory center • Level of consciousness (ascending reticular formation) • Ascending sensory tracts • Descending motor tracts	Posterior	PICA VA ASA	• Dysphagia *Not represented on NIHSS* • Tongue deviation *Not represented on NIHSS* • Decreased LOC *NIHSS Item: 1* • Cardiopulmonary instability *Not represented on NIHSS* • Vasomotor instability *Not represented on NIHSS* • Sensory/motor pathway disruption *NIHSS Items: 4,5,6,8*

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Table 2
National Institutes of Health Stroke Scale

1a: LOC	0 = Alert 1 = Arousable by mild stimulation 2 = Obtunded/arousable to pain 3 = Unresponsive/reflex only
1b: Questions (month and age)	0 = Answers both correctly 1 = Answers one correctly 2 = None are correct
1c: Follows commands	0 = Performs both correctly 1 = Performs one correctly 2 = Performs neither correctly
2: Best gaze	0 = Normal 1 = Partial deviation 2 = Forced deviation
3: Visual fields	0 = No visual loss 1 = Partial hemianopia 2 = Complete hemianopia 3 = Bliateral hemianopia/blind
4: Facial palsy	0 = Normal 1 = Minor 2 = Partial 3 = Complete
5a: Motor left arm	0 = No drift 1 = Drift 2 = Some effort against gravity 3 = No effort against gravity 4 = No movement
5b: Motor right arm	0 = No drift 1 = Drift 2 = Some effort against gravity 3 = No effort against gravity 4 = No movement
6a: Motor left leg	0 = No drift 1 = Drift 2 = Some effort against gravity 3 = No effort against gravity 4 = No movement
6b: Motor right leg	0 = No drift 1 = Drift 2 = Some effort against gravity 3 = No effort against gravity 4 = No movement
7: Limb ataxia	0 = Absent 1 = Present in 1 limb 2 = Present in 2 limbs UN = Untestable
8: Sensory	0 = Normal 1 = Mild to moderate loss 2 = Severe to total loss UN = Untestable

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Table 2
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9: Best language	0 = No aphasia 1 = Mild to moderate aphasia 2 = Severe aphasia 3 = Mute, global aphasia
10: Dysarthria	0 = No abnormality 1 = Mild to moderate 2 = Severe UN = Intubated or physical barrier
11: Extinction/inattention (neglect)	0 = No abnormality 1 = Extinction to 1 modality 2 = Extinction to >1 modality

that 7 total points require intact language skills, so that left MCA stroke patients generally will have much higher NIHSS scores than patients with right MCA territory strokes of equal infarct size.[10] Despite this bias, the NIHSS is widely acknowledged to be the gold standard[6] for ischemic stroke patient assessment, and therefore, mastering use of this instrument is a prerequisite for acute stroke nurses.[11] Although the Glasgow Coma Scale (GCS) has been widely used as a neurologic assessment measure, the scale was originally developed as a level-of-consciousness (LOC) measure to prognosticate outcome in patients with traumatic brain injury.[12,13] In addition, GCS scores require recording the patient's "best response," so that in a stroke patient with normal LOC, the disabled side would not be scored if the tool were used correctly; in other words, despite being highly disabled, it is likely that a GCS may correctly score a patient as normal (score of 15).[13] Use of modified versions or what have been called "slim" versions of the NIHSS has been shown to be unacceptable by stroke nurses in that these scales may not capture the arterial territory affected by stroke, thereby giving false reassurance that a patient is not disabled when disability actually exists.[13] Therefore, the evidence-based recommendation widely accepted today is use of the full NIHSS, with use of just the NIHSS elements that are positive for the disability used during more frequent assessments for patients treated with alteplase (tissue plasminogen activator).[13,14] Recently, the NIHSS has been validated for use in patients with intraparenchymal hemorrhage.[15]

Level of consciousness (National Institutes of Health Stroke Scale item 1)

Evaluation of LOC involves determining the degree of wakefulness in a patient and makes up one of the most important elements of the neurologic examination because it is a sensitive indicator of neurologic change.[16] Descriptors commonly used for LOC include alert (wide awake), lethargic (sleepy, dull, indifferent), obtunded (deep sleep; difficult to arouse), stupor (unarousable, yet with normal response [withdrawal or localization] to noxious stimuli), or comatose (unarousable with abnormal response [mass response or absent response] to stimuli).[17,18] Evaluation of LOC is included in NIHSS items 1a (wakefulness measure), 1b (responsiveness to questions), and 1c (responsiveness to commands). Generally, patients suffering anterior circulation ischemic stroke will have a normal LOC with the exception of catastrophic bilateral anterior circulation lesions[19]; this may remain true throughout the course of stroke completion with the exception of larger MCA stroke that is subject to severe or even malignant edema, which sets in and worsens within 24 to 28 hours after completed infarction.[20] Posterior circulation strokes are generally at greater risk of LOC deterioration, in particular, brainstem strokes, with disruption of ascending reticular formation and

those affecting the cerebellum, which may cause brainstem compression and ventricular system obstruction.[1]

Cranial nerve examination on the National Institutes of Health Stroke Scale

There are 12 pairs of cranial nerves (CN) that are generally classified as part of the peripheral nervous system; however, this is anatomically inaccurate in that CN II originates anatomically from the central nervous system.[4] A detailed description of CN anatomy and assessment is an article in and of itself, but suffice it to say that CNs I and III–XII are considered true peripheral nerves, whereas CN II (optic nerve) is derived from the optic stalks of the diencephalon composed of retinal ganglion cell axons and glial cells, and similar to other axons of the central nervous system, covered with myelin. Assessment of CN I (olfactory), which originates in the forebrain, is generally deferred, whereas assessments of CNs III–VII, which have their cell bodies in the brainstem, are incorporated within the NIHSS (see **Table 1**) with the exception of pupillary constriction to light, which can be added in the case of altered LOC or findings consistent with palsies of CNs affecting extraocular movement (EOM; CNs III, IV, and VI).[1,4] CNs IX, X, and XII make up important components of stroke patient assessment in that swallow integrity is insulted in about 50% of stroke patients[21,22]; swallow assessment will not be discussed in this article but often involves a comprehensive battery of tests starting with a clinical bedside screen, assessment of LOC which must be assessed to ensure adequate intake and airway protection, and fiber-optic endoscopic evaluation of swallow completed by a speech and language pathologist.[23] In addition, insults to CNs VII, IX, and/or XII may also impact speech articulation, resulting in a dysarthric quality that may significantly limit communication. Assessment of CN XI (spinal accessory) can be easily incorporated into a neurologic examination through the addition of shoulder shrug and head turning.[24] CN VIII (vestibulocochlear) is rarely tested unless the patient presents with stroke symptoms accompanied by a sudden-onset acute hearing loss or acute onset of vertigo; when this is the case, suspicion for brainstem infarction should remain high, especially when accompanied by other brainstem findings, although temporal lobe infarction may be a consideration.[25]

Extraocular movement (National Institutes of Health Stroke Scale item 2)

Assessment of extraocular movements (EOM) provides the examiner with information reflective of brainstem integrity and level of brainstem dysfunction given that the cell bodies of these nerves originate in the midbrain (CN III and IV) and pons (CN VI).[26] Classically, the patient is asked to move his or her eyes to trace the movement of the examiner's fingers, generally in an "H" configuration, whereas the NIHSS asks the patient to trace the examiner's fingers through lateral eye movement.[19] In the case of altered LOC, oculocephalic maneuver or doll's eyes are examined, and an absence of full movement would indicate brainstem dysfunction.[27] In the conscious patient who is unable to understand what the examiner requests, the examiner should move about the patient to draw eye movement in his or her direction so EOMs can be evaluated.[19]

Findings from EOM testing must be differentiated from findings associated with the frontal eye fields (see **Fig. 1**; between Brodmann areas 4, 6, and 8, by the middle frontal and precentral gyri). The frontal eye fields are responsible for saccadic eye movements that normally allow the eyes to rapidly move together to surveil visual content; interruption of this area of the brain causes conjugate eye deviation toward the affected MCA territory.[28] In contrast, brainstem CN III, IV, and VI findings would include either lateral deviation, ptosis, and pupillary dilation (CN III;

oculomotor), loss of superior oblique innervation with upward migration and vertical diplopia that worsens when attempting to look down (CN IV; trochlear), or midline eye deviation with limited lateral gaze producing diplopia (CN VI; abducens).[27] With significant brainstem destruction, patients may present with a fixed gaze in combination with other classic brainstem findings, such as decreased LOC, including coma.[29]

Peripheral vision (National Institutes of Health Stroke Scale item 3)

Peripheral vision (CN II) is assessed by evaluating each eye independently and both eyes collectively.[19] Vision for each eye is divided into 4 quadrants, and visual loss is determined to be either monocular (resulting from prechiasmal stroke affecting the retina or optic nerves) or binocular (stroke at or posterior to the chiasm).[1,19] The more posterior along the visual tracts that the stroke occurs, the more congruent the deficit is in each eye.[1,30] Visual field loss can be affected by the OA (monocular), MCA (binocular), PCA (binocular), or top of the BA (binocular, including bilateral hemianopia, also referred to as cortical blindness) stroke.[29] In the case of MCA stroke producing visual loss, the examiner should expect vision loss to also be accompanied by findings classically associated with MCA stroke, such as voluntary motor and/or primary sensory loss, aphasia (usually left hemisphere), and/or neglect (usually right hemisphere),[29,31] whereas in pure P2 PCA stroke, the examiner would expect binocular hemianopia without other disability findings.[32]

To evaluate visual fields, the examiner assesses the patient's ability to visualize an object (usually the examiner's fingers) in each of the 4 quadrants of vision.[19] Although some NIHSS training films demonstrate this assessment by asking the patient to state how many fingers he or she is able to see in each field, this often leads to confusion. For example, if the patient says "2 fingers" when 3 fingers were actually held up, some examiners miss-score this as visual field loss; however, the patient did in fact see the examiner's fingers, so vision should be assumed to be intact. The error in this example may actually be on the part of the examiner through failure to accurately position fingers within the patient's visual field. Therefore, it is often clearer to just simply have the patient state "now" when he or she sees the examiner's finger start wiggling in each quadrant. However, by far the most common error made by novice examiners is holding their arms/fingers too far into the periphery for the patient to see. Examiners must remember to use themselves as confirmation and make sure that their fingers are held well within a field of vision that is easily seen to avoid assuming visual loss in a patient who has normal visual fields. For patients unable to participate in this examination because of altered LOC, cognitive or language dysfunction, blinking-to-threat will suffice as confirmation of intact vision.[28]

Facial symmetry (National Institutes of Health Stroke Scale item 4)

Nervous control of facial expression arises from the corticobulbar fibers that originate on the lower lateral side of the precentral gyrus (see **Fig. 1**; Brodmann area 4; MCA territory), coursing through the corona radiata, internal capsule, and medial cerebral peduncle to the pons where most fibers decussate and then terminate in the contralateral facial motor cell body.[1] Stroke affecting the MCA territory interferes with precentral gyrus transmission, producing what is often referred to as "central VII" motor loss, whereas brainstem stroke at the site of the pontine CN VII nucleus results in loss of peripheral facial innervation. With central VII loss, the upper face is commonly spared, and there are several physiologic theories explaining this preserved function, including bilateral supranuclear control and the scant direct cortical

innervation provided to the upper face. Therefore, loss of the upper face is expected to occur most commonly with brainstem pontine stroke and is considered uncommon in MCA distribution stroke.[1,33] Assessment of symmetric facial expression (CN VII) is assessed by asking the patient to smile, blow out the cheeks, close the eyes tightly, and raise the eyebrows.[19] Dysarthria or slurred speech (NIHSS item 10) is commonly found in relation to motor loss affecting the lower face.[19] Expected findings align with vascular territories as follows:

- MCA stroke: flattening of the nasolabial folds on the same side as limb weakness[1]
- BA distribution stroke:[1]
  - Upper brainstem, coextensive (face, arm, leg) contralateral weakness in relation to the infarction
  - Lower pontine, facial droop ipsilateral to the pontine infarction with limb weakness contralateral to pontine infarction.

With extremely rare exceptions caused by discreet medullary stroke lesions (Dejerine anterior bulbar[1,34] [medial medullary] syndrome or pure Wallenberg [lateral medullary] syndrome that excludes Opalski[35] [submedullary] components), examiners should be highly suspect of presentations involving motor loss of the limbs with sparing of the face because this presentation may suggest a stroke mimic diagnosis. Completion of emergent MRI, diffusion-weighted imaging/apparent diffusion coefficient, and fluid attenuated inversion recovery sequences will immediately confirm or dispel stroke diagnosis with such a presentation.[36]

Motor Examination on the National Institutes of Health Stroke Scale

The motor homunculus, Latin for "little man" (**Fig. 4**), is a graphic depiction of what a man would look like if the body parts were proportionate to the amount/size of brain territory on the precentral gyrus (see **Fig. 1**; Brodmann area 4) dedicated to specific types of motor function. The most superior and medial areas of the precentral gyrus

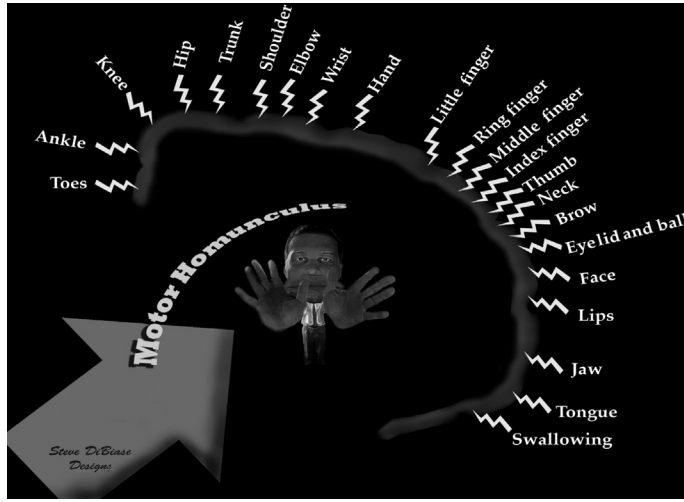


Fig. 4. Motor homunculus. (*Reprinted* with permission, Health Outcomes Institute, LLC, Fountain Hills, Arizona, USA.)

control voluntary movement for the lower extremities; progressing laterally and inferiorly, the homunculus distribution includes the trunk, arm, hand, face, lips, tongue, and larynx. When considering the layout of the motor homunculus in relation to the anterior circulation's distribution, the ACA supplies blood flow to the areas of the precentral gyrus that innervate voluntary motor function in the lower limbs, whereas MCA distribution supplies blood flow to areas on the gyrus responsible for innervation of the arms and facial and orolingual structures. For this reason, MCA distribution strokes commonly present with similar motor loss in the face and arm; the more distal the MCA occlusion, leg involvement may be entirely spared, whereas in distal ICA/proximal MCA stroke near in proximity to the ACA A1 division (see **Fig. 2**), patients may present with face, arm, and leg motor loss.[1]

A large portion of the corticospinal tract originates in the precentral gyrus and descends through the internal capsule into the brainstem where most motor fibers decussate in the pyramids of the medulla before descending to the spinal cord and lower motor neurons. In this way, the right and left precentral gyri control voluntary motor function for the opposite sides of the body. Stroke that interferes with the descent and distribution of the corticospinal tract within the cortex, subcortex, or brainstem will cause voluntary limb motor loss on the opposite side of the body; Opalski syndrome complicating a Wallenberg presentation is a rare exception in that motor extremity loss is ipsilateral to the medullary stroke.[1]

Motor function is classically graded using a muscle strength scale of 0 to 5, where 0 equals no movement and 5 equals normal power (**Box 1**)[1]; findings are presented as a fraction with the patient's actual muscle strength in the numerator and normal muscle strength (grade 5) in the denominator. Motor function is also measured on the NIHSS in item 5 (motor arm) and item 6 (motor leg)[19,37]; the number assigned on the NIHSS should not be confused with the motor grade in that the NIHSS is a scale of which numeric values add weight to connotative severity on an ordinal scale.

Ataxia is a disturbance of smooth performance of voluntary motor function, resulting in errors in movement timing, force, range, and speed.[38] Stroke affecting the cerebellum results in motor function without inhibition and modulation, producing inaccurate and inconsistent movement.[39] Ataxia is tested on the NIHSS (item 7) using the finger-nose-finger, and heel-shin tests. The examiner has the patient reach using the full length of his or her arm to touch the examiner's finger; the examiner can move placement of his or her finger with each subsequent reach by the patient, while observing for smooth, accurate movement. The heel-shin test is performed with the patient lying on his or her back. The patient is asked to run his or her heel down the shin of the opposite leg and back up again to the knee; the test is repeated in both limbs while the examiner observes for smooth, accurate limb motion.[19] Misinterpretation of ataxia findings is common. When

Box 1
Muscle strength grading

0 = No muscle activation

1 = Trace muscle activation, such as a twitch, without achieving full range of motion

2 = Muscle activation with gravity eliminated, achieving full range of motion

3 = Muscle activation against gravity, full range of motion

4 = Muscle activation against some resistance, full range of motion

5 = Muscle activation against the examiner's full resistance, full range of motion

motor weakness results in loss of coordination, ataxia should be scored as a 0 on the NIHSS; however, when motor strength is normal, but movement lacks coordination, ataxia should be scored as present.[19]

Sensory Examination

Sensory pathways ascend the spinal cord decussating within either the cord or the brainstem before reaching the thalamus where they are relayed to the parietal lobe's postcentral gyrus.[1] The postcentral gyrus (see **Fig. 1**; Brodmann areas 1–3) houses the primary somatosensory cortex and can be illustrated with a sensory homunculus of which the layout is distributed by the amount of gyrus brain tissue dedicated to various parts of the body.[1]

Proprioception has a conscious and an unconscious component. The conscious pathway connects with the thalamus and the cerebral cortex, enabling an understanding of the position of the body in space, whereas the unconscious pathway connects with the cerebellum, facilitating walking and complex acts without having to think about which joints to flex and or extend. Although proprioception is not included as part of the NIHSS, it can be easily added to the neurologic examination when other sensory findings are identified; the test is performed with the patient's eyes closed, and the examiner then moves the patient's finger up or down, right or left, asking the patient what direction the joint is moving in.[1,40]

Examination of primary sensory function on the NIHSS includes testing of touch sensation on the face, arms, and legs. The examiner begins by touching the side that appears unaffected by stroke with a pin and asking the patient if it feels "sharp"; the examiner then moves to the affected side, repeating the procedure and asking the patient if it feels "as sharp" as the unaffected side. It is important to use a pin so that the patient can clearly discriminate the sensation, because use of touch alone may mask an actual sensory loss.[19]

In general, the right hemisphere dominates in completion of tasks that require nonvisual spatial attention, especially in right-handed persons.[41–43] Attention activated by external stimuli is localized to the inferior parietal lobe and the posterior aspect of the superior temporal gyrus with the right hemisphere alerting both sides of the body[1]; because of this, hemineglect syndromes are most commonly associated with right parietal MCA lesions.[44]

The NIHSS assessment of neglect (item 11) includes a tactile and a visuospatial component. Tactile neglect is tested with the patient's eyes closed to determine if the patient can distinguish being touched simultaneously on each side; in the case of left hemineglect syndrome, the patient is unable to consistently recognize simultaneous touch on the left side of his or her body. The examiner touches 1 side of the face or both sides of the face simultaneously with a brief on-off low-pressure movement and asks the patient to state which side was touched; the procedure is then repeated in the patient's arms and, if necessary, legs.[19] Visuospatial neglect is tested during assessment of NIHSS item 3 (to avoid redundant eye testing) in patients with intact visual fields, and findings are incorporated into the scoring of neglect on item 11. The examiner positions his or her fingers within the visual fields of both the patient's eyes; most commonly, the examiner asks the patient to state which finger is wiggling, right, left, or both, starting first by wiggling a finger on 1 hand, then the other hand, and then providing double simultaneous stimuli by wiggling a finger on both hands at the same time. Patients with hemineglect may present with only tactile or visuospatial neglect, or they may present with profound neglect, whereby both tactile and visuospatial attention are affected.[1,19,45]

Language

Language is considered a sensorimotor function of the brain in that it requires functions tied to receiving and processing language (sensory) as well as language expression (motor).[1] In general, the left hemisphere of the brain houses language and linguistic functions, along with other executive functions, such as mathematical skills, reasoning, and analysis.[1] Aphasia is the inability to comprehend or formulate language[46] and is tested by NIHSS item 9.[19] Motor language aphasia produces expressive loss because of frontal cortical MCA stroke in the Broca area[47] (see **Fig. 1**; Brodmann area 44) and is tested through examination of language fluency on the NIHSS by asking the patient to name items and describe a scene.[19,48] Sensory language aphasia produces an inability to receive language because of MCA stroke in Wernicke area (see **Fig. 1**; Brodmann area 37, 39, and 40), resulting in an inability to assemble phonemes accurately, an inability to repeat sentences correctly, and an inability to name items correctly.[19,49,50] However, compared with those with Broca aphasia, patients with Wernicke aphasia have effortless, yet difficult-to-decipher speech, and therefore, this type of disability is often described as a "fluent aphasia."[51] The NIHSS item 9 assesses receptive deficits by examining language fluency as the patient describes pictures and reads sentences; the examiner can also attempt to engage the patient in conversation to further differentiate aphasia type.[19] Global aphasia presentations are associated with large left MCA stroke, often presenting with perisylvian lesions, that produce both Wernicke and Broca aphasia components, leaving the patient mute with impaired comprehension.[46]

SUMMARY

A thorough and precise neurologic examination is essential to rapid diagnosis, treatment, and ongoing evaluation of acute stroke patients. Although challenging, practice and repetition, alongside a commitment to furthering knowledge of neurologic anatomy and physiology, are the keys to examiner proficiency and expertise. Subtle neurologic changes requiring immediate intervention often can only be identified through completion of serial assessments. Because stroke evolves over minutes to even days, nurses caring for acute stroke patients must seek to master assessment skills with an eye to the vascular territory affected to reduce significant risks for devastating disability, deterioration, and death.

DISCLOSURE

The authors have nothing to disclose.

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